

Module 6 _ Application of Natural Language Processing

MACHINE TRANSLATION

1. Explanation

Machine Translation is a subfield of **Natural Language Processing (NLP)** that deals with the **automatic translation of text or speech** from one language to another using computers.

It aims to make communication easier between people who speak different languages — for example, translating English text into Hindi or French automatically.

MT systems analyze the **grammar, semantics, and structure** of both source and target languages to produce accurate and meaningful translations.

Well-known examples include **Google Translate, Microsoft Translator, and DeepL.**

2. Process of Machine Translation

The process of MT generally involves the following main stages:

1. Analysis (Source Language Processing)

- The system reads and analyzes the structure of the source language text.
- It performs **tokenization, part-of-speech tagging, syntactic parsing, and semantic analysis** to understand the meaning.

2. Transfer (Mapping Between Languages)

- The analyzed information from the source language is **converted** into a representation suitable for the target language.
- This step involves **bilingual dictionaries and grammar rules** to align meaning.

3. Generation (Target Language Processing)

- The system generates text in the target language that conveys the same meaning.
- It ensures grammatical correctness and naturalness of the translated output.

3. Types of Machine Translation

There are **three main types** of Machine Translation systems:

1. Rule-Based Machine Translation (RBMT)

- Uses a large set of **linguistic rules** and **bilingual dictionaries**.
- Translation is based on grammatical and syntactic structures.
- *Advantage:* Good for grammatically controlled output.
- *Limitation:* Requires extensive linguistic knowledge and is hard to maintain.

2. Statistical Machine Translation (SMT)

- Uses **probabilistic models** learned from large bilingual text corpora.

- Translations are based on the **statistical likelihood** of word or phrase alignments.
- Example: *Google Translate (before 2016)*.
- *Advantage*: Learns automatically from data.
- *Limitation*: May produce grammatically awkward sentences.

3. Neural Machine Translation (NMT)

- Uses **deep neural networks** (especially encoder-decoder models with attention or transformers).
- Translates entire sentences rather than word-by-word.
- Example: *Modern Google Translate, DeepL*.
- *Advantage*: Produces fluent, context-aware, and accurate translations.
- *Limitation*: Requires large training data and computational power.

Steps in Machine Translation Process

1. Text Analysis (Source Language Processing)

- The system first analyzes the input sentence in the **source language**.
- It performs:
 - **Tokenization**: Splitting text into words or phrases.
 - **Part-of-Speech Tagging**: Identifying nouns, verbs, adjectives, etc.
 - **Syntactic Analysis (Parsing)**: Understanding sentence structure.
 - **Semantic Analysis**: Interpreting the meaning of words and phrases.

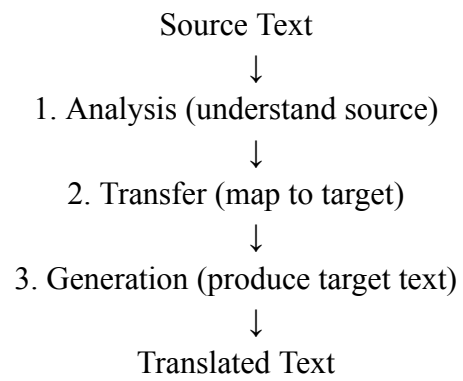
2. Transfer (Conversion to Target Language Structure)

- The analyzed source text is **mapped** to the equivalent structure in the **target language**.
- This includes:
 - Translating words using **bilingual dictionaries** or **learned models**.
 - Adjusting **word order** and **grammar** to fit the target language rules.

3. Generation (Target Language Processing)

- The system **generates** the translated sentence in the target language.
- It ensures that:
 - The sentence is **grammatically correct**,
 - The translation is **natural and fluent**,
 - The **meaning remains the same** as the original text.

⚙️ Simplified Diagram of the Process



TEXT SUMMARIZATION

1. Explanation

Text Summarization is a technique in **Natural Language Processing (NLP)** that aims to **shorten large pieces of text** while **preserving the main meaning and key information**.

It helps users quickly understand the **core ideas** of a document without reading the entire text.

This process is widely used in applications such as:

- News summarization
- Research paper abstracts
- Email and legal document condensation
- Social media monitoring

The goal is to produce a **concise, coherent summary** that captures the essence of the original content.

2. Types of Text Summarization

There are **two main types** of text summarization techniques:

A. Extractive Summarization

- This approach **selects important sentences, phrases, or words** directly from the original text.
- It does **not generate new sentences**, but picks the most relevant ones and combines them into a summary.
- It relies on **ranking algorithms, word frequency, or statistical measures** (like TF-IDF, TextRank).

Example:

Original text:

“Machine Translation is a process of automatically converting text from one language to another. It helps break language barriers.”

Extractive summary:

“Machine Translation automatically converts text from one language to another.”

Advantages:

- Simple and fast
- Maintains grammatical correctness (as sentences are from original text)

Limitations:

- May lack coherence
- Doesn't rephrase or interpret meaning

B. Abstractive Summarization

- This approach **understands the meaning** of the original text and **generates new sentences** to express the same idea.
- It works like how humans write summaries — rephrasing and compressing the content.
- Uses **Deep Learning models** such as **Seq2Seq, Transformer, BERT, or GPT-based** architectures.

Example:

Original text:

“Machine Translation is a process of automatically converting text from one language to another. It helps break language barriers.”

Abstractive summary:

“Machine Translation enables automatic cross-language communication.”

Advantages:

- More natural and human-like
- Can paraphrase and generalize information

Limitations:

- Computationally expensive
- Requires large training datasets

Steps in Text Summarization

The process of text summarization — whether extractive or abstractive — generally follows **five major steps**:

1. Text Preprocessing

- The first step is to **clean and prepare the input text** so that the model can understand it properly.
- This includes:
 - Removing punctuation, numbers, and stop words (like *a*, *an*, *the*)
 - Converting text to lowercase
 - Tokenizing (splitting text into words or sentences)
 - Lemmatization or stemming (reducing words to their root form)

Example:

Original: “Machine Translation helps people communicate across languages.”

After preprocessing → [machine, translation, help, people, communicate, across, language]

2. Text Representation

- The cleaned text is then **converted into a numerical or vector form** that the computer can process.
- Common techniques include:
 - **Bag of Words (BoW)**
 - **TF-IDF (Term Frequency–Inverse Document Frequency)**
 - **Word Embeddings (Word2Vec, GloVe, BERT embeddings)**

3. Sentence Scoring or Understanding

Depending on the method used:

• For Extractive Summarization:

- Each sentence is **scored** based on importance using:
 - Word frequency
 - Sentence position (first/last often important)
 - Similarity measures
 - Graph-based ranking (e.g., **TextRank Algorithm**)

• For Abstractive Summarization:

- A **deep learning model** (like a **Seq2Seq** or **Transformer**) learns to understand the **meaning** of the text and encodes it into a **semantic representation**.

4. Summary Generation

- Based on the sentence scores or the semantic understanding:
 - **Extractive models** → pick top-ranked sentences.
 - **Abstractive models** → generate new sentences using the learned context.

Example:

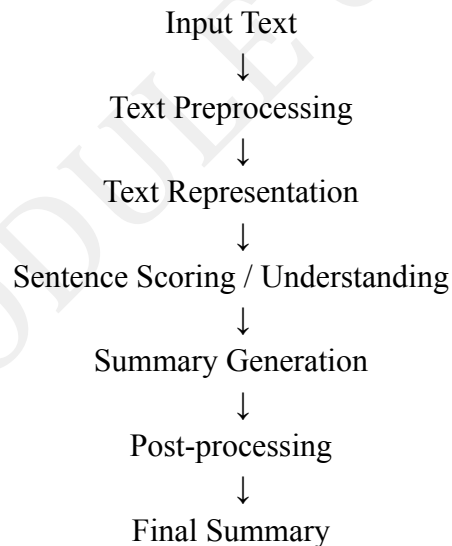
Original paragraph (5 lines) → Summary (1 or 2 key lines)

5. Post-processing (Refinement)

- The generated summary is refined to ensure:
 - Grammatical correctness
 - Coherence (smooth flow between sentences)
 - Elimination of redundancy
 - Proper punctuation and capitalization

This step produces the **final readable summary**.

Simplified Workflow Diagram



SENTIMENT ANALYSIS

1. Explanation

Sentiment Analysis (also known as **Opinion Mining**) is a technique in **Natural Language Processing (NLP)** used to **identify and classify the emotional tone** or **sentiment** expressed in a piece of text.

It determines whether the writer's attitude toward a topic, product, or service is:

- **Positive**,
- **Negative**, or
- **Neutral**.

Sentiment analysis helps computers understand **human emotions and opinions** from text data such as reviews, comments, tweets, or feedback.

It uses **machine learning**, **deep learning**, and **lexicon-based** approaches to analyze linguistic features like words, phrases, and context.

2. Example

Example 1 – Product Review:

Text: *"The camera quality is amazing, but the battery drains too fast."*

→ Sentiment: **Mixed (Positive for camera, Negative for battery)**

Example 2 – Movie Review:

Text: *"The movie was fantastic and full of emotions."*

→ Sentiment: **Positive**

Example 3 – Tweet:

Text: *"I'm so disappointed with the new update."*

→ Sentiment: **Negative**

3. Applications of Sentiment Analysis

Area	Use Case
Business & Marketing	Analyzing customer feedback, brand reputation, and market trends

Social Media Monitoring	Tracking public opinion on platforms like Twitter, Facebook, etc.
Product Reviews	Understanding customer satisfaction on e-commerce websites
Politics	Measuring public opinion about leaders or policies
Customer Support	Detecting frustrated or unhappy customers automatically
Healthcare	Analyzing patient feedback and mental health through text patterns

Steps in Sentiment Analysis

Sentiment Analysis generally follows **five key steps** — from preparing the text to identifying whether it’s positive, negative, or neutral.

Data Collection

- The first step is to **gather text data** from different sources such as:
 - Product or movie reviews
 - Tweets and social media posts
 - Customer feedback or surveys
- This collected text forms the **dataset** on which analysis will be performed.

Example:

Gather 1,000 tweets about a new smartphone release.

Text Preprocessing

- The raw text is cleaned and prepared for analysis.
- This involves:
 - Removing punctuation, emojis, and special symbols
 - Converting text to lowercase
 - Removing stop words (e.g., *is, the, a, an*)
 - Tokenization (splitting sentences into words)
 - Lemmatization/Stemming (reducing words to root form)

Example:

“Phones are AMAZING!!!” → “phone amazing”

Feature Extraction (Text Representation)

- The text is **converted into numerical form** so the machine can process it.
- Common methods include:
 - **Bag of Words (BoW)**
 - **TF-IDF (Term Frequency–Inverse Document Frequency)**
 - **Word Embeddings** (Word2Vec, GloVe, BERT)

These features represent how important each word is in the sentence or document.

Sentiment Classification

- A **machine learning or deep learning model** is applied to classify the sentiment as:
 - **Positive**
 - **Negative**
 - **Neutral**

Techniques used:

- **Machine Learning:** Naïve Bayes, SVM, Logistic Regression
- **Deep Learning:** LSTM, CNN, Transformer, BERT

Example:

Input: “The movie was boring.”

→ Model Output: **Negative**

Output Interpretation / Visualization

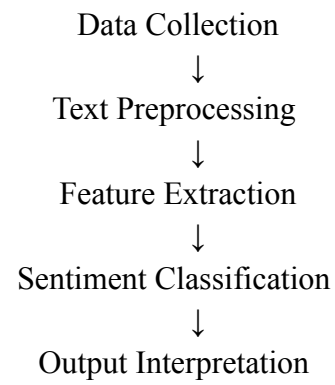
- The system presents the final results — often as:
 - **Text labels** (Positive, Negative, Neutral)
 - **Polarity scores** (e.g., +0.85 = strong positive)
 - **Graphs or dashboards** for visual analysis

Example:

Out of 1000 tweets:

- 700 Positive
- 200 Negative
- 100 Neutral

Simplified Workflow Diagram



INFORMATION RETRIEVAL

1. Explanation

Information Retrieval (IR) is a branch of **Natural Language Processing (NLP)** and **Computer Science** that deals with **finding and retrieving relevant information** from large collections of data — such as documents, web pages, or databases — **in response to a user's query**.

In simple words, **IR systems help users find what they are looking for** from a massive amount of unstructured text data (like on the internet).

A common example of an IR system is a **search engine** such as **Google, Bing, or Yahoo**, which retrieves the most relevant web pages for the query you type.

Example:

User query: “Symptoms of diabetes”

→ IR system retrieves web pages, research papers, and articles related to diabetes symptoms.

2. Process of Information Retrieval

The **Information Retrieval process** typically consists of several key steps that help locate, rank, and display relevant information:

Step 1: Document Collection

- The system gathers a large set of documents from different sources such as websites, books, databases, or reports.
- This collection forms the **document repository** or **corpus**.

Step 2: Text Preprocessing (Indexing)

- The collected documents are **preprocessed** and **indexed** for faster searching.
- Preprocessing includes:
 - Tokenization (splitting text into words)
 - Stop-word removal (removing common words like “the”, “is”)
 - Stemming or Lemmatization (reducing words to their base form)
- The system creates an **inverted index** — a data structure that maps **words to the documents** they appear in.

Example:

Word: “Diabetes” → Appears in Docs [3, 7, 15, 18]

Step 3: Query Processing

- When a user types a search query, it is also **processed and transformed** (tokenized, stemmed, etc.) in the same way as documents.
- This ensures the system can compare the query to indexed documents accurately.

Example:

Query “treatment of diabetes” → Tokens: [treatment, diabetes]

Step 4: Document Retrieval and Matching

- The system compares the **processed query** with the **indexed documents** to find matches.
- It uses **similarity measures** such as:
 - **Cosine similarity**
 - **TF-IDF weighting**
 - **BM25 ranking function**

Each document is given a **relevance score** based on how closely it matches the query.

Step 5: Ranking and Output Display

- The matched documents are **ranked in descending order of relevance**.
- The top-ranked results are displayed to the user (like the first few search results on Google).

Example:

Query: “machine translation”

Top results:

Research paper on MT

Wikipedia article

Blog on neural MT

3. Types of Information Retrieval Systems

Information Retrieval systems can be classified into **three main types** based on how they match documents and queries:

A. Boolean Retrieval Model

- Uses **logical operators** such as **AND, OR, NOT** to match documents.
- Only retrieves documents that **exactly satisfy** the query conditions.

Example:

Query: “*machine AND translation*” → returns only documents containing both words.

Advantage: Simple and fast.

Limitation: Does not rank results by relevance.

B. Vector Space Model

- Represents documents and queries as **vectors** in multi-dimensional space.
- Measures similarity using metrics like **cosine similarity**.
- Ranks documents by **degree of relevance** rather than exact matching.

Advantage: Supports partial matching and ranking.

Limitation: Computationally heavier than Boolean model.

C. Probabilistic Model

- Estimates the **probability that a document is relevant** to a given query.
- Uses models such as **BM25** or **Bayesian inference**.
- Modern search engines use probabilistic and learning-based versions of this model.

Advantage: More accurate and adaptive to user behavior.

Limitation: Complex to implement and train.

Applications of Information Retrieval

- Web Search Engines (Google, Bing, Yahoo)
- Digital Libraries
- Question Answering Systems
- Recommendation Systems
- Enterprise Document Management

Steps in Information Retrieval

Information Retrieval systems follow a **systematic process** to find and display the most relevant documents for a user's query.

The main steps are:

Document Collection

- The first step is to **gather a large set of documents** (data sources) from which information will be retrieved.
- Documents may come from:
 - Web pages
 - Books and journals
 - Databases

- Reports, news, etc.
- These documents together form a **corpus** (the entire searchable collection).

Example:

Collecting all research papers related to “Artificial Intelligence”.

Text Preprocessing

- Before retrieval, all documents must be **cleaned and standardized**.
- This step improves efficiency and accuracy.
- It includes:
 - **Tokenization:** Splitting text into words or terms
 - **Stop-word Removal:** Removing common words like *a, an, the*
 - **Stemming/Lemmatization:** Converting words to their root form
 - **Lowercasing and Removing Punctuation**

Example:

“Retrieving Information Systems” → [retrieve, information, system]

Indexing

- After preprocessing, documents are **indexed** for fast searching.
- The system creates an **inverted index**, which maps **each word to the list of documents** where it occurs.
- This helps retrieve documents quickly when a query is entered.

Example:

Word “Translation” → Appears in Docs [2, 7, 15, 21]

Query Processing

- When a user types a query, it is also **processed in the same way** as the documents (tokenized, stemmed, etc.).
- This ensures that both query and document terms are in the same form for accurate comparison.

Example:

Query: “Machine learning applications” → [machine, learn, application]

Document Retrieval / Matching

- The system **compares the processed query** with the **indexed documents**.
- It identifies documents that contain matching or similar terms.
- Techniques used:
 - **Boolean Matching (AND, OR, NOT)**
 - **Vector Space Model (Cosine Similarity)**
 - **Probabilistic Ranking (BM25)**

Example:

Query: “Neural network translation” → Documents 5, 8, and 12 are relevant.

Ranking and Scoring

- Retrieved documents are **ranked** based on **relevance scores** (how well they match the query).
- The most relevant ones are displayed first (like top search results on Google).
- Scoring methods include:
 - **TF-IDF**
 - **Cosine Similarity**
 - **BM25**
 - **Learning to Rank algorithms**

Result Display

- Finally, the system **displays the top-ranked documents** to the user, usually with:
 - Title, snippet, and source link
 - Option to view or refine the query

Example:

Query: “Machine Translation”

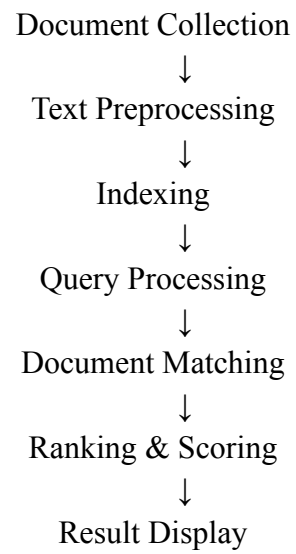
Top Results:

Wikipedia article

Research paper

Blog on Neural MT

Simplified Workflow Diagram



QUESTION ANSWERING SYSTEM

1. Explanation

A **Question Answering System** is an **Artificial Intelligence (AI)** and **Natural Language Processing (NLP)** system designed to **automatically answer questions posed by humans in natural language**.

Unlike simple search engines that return a list of documents, a QAS aims to provide a **precise and concise answer**.

Key features:

- Understands the **query semantics** (what the user is asking)
- Searches the **knowledge base or documents** for relevant information
- Generates the **most accurate answer** in natural language
- *Example:*
- User Query: “Who is the CEO of Tesla?”
- QAS Output: “Elon Musk”

2. Steps in a Question Answering System

The process of QAS involves **several key steps**:

Step 1: Question Analysis

- The system first **analyzes the question** to understand its:
 - **Type:** Fact-based, list, definition, yes/no, etc.
 - **Keywords:** Important words in the query
 - **Expected answer type:** Person, date, location, number, etc.

Example:

Query: “When was Google founded?”

- Keywords: [Google, founded]
- Expected Answer Type: Date

Step 2: Document or Knowledge Base Retrieval

- The system **searches a database, corpus, or web** to retrieve **relevant documents or passages** that may contain the answer.

- Techniques used:
 - Information Retrieval (IR) methods
 - Search engines or knowledge graphs

Example:

Query: “When was Google founded?”

- Retrieved document: “Google was founded on September 4, 1998, by Larry Page and Sergey Brin.”

Step 3: Passage or Candidate Answer Extraction

- From the retrieved documents, the system **identifies possible answer sentences or phrases**.
- Methods used:
 - Pattern matching (e.g., regex for dates or names)
 - Named Entity Recognition (NER)
 - Deep learning-based models

Example:

Candidate answer extracted: “September 4, 1998”

Step 4: Answer Selection / Ranking

- Multiple candidate answers are **ranked based on relevance and confidence**.
- The system selects the **best matching answer**.
- Modern systems use **neural models** like BERT, GPT, or Transformers for higher accuracy.

Example:

Candidate answers:

“September 4, 1998”

“Founded by Larry Page”

→ Selected: “September 4, 1998”

Step 5: Answer Presentation

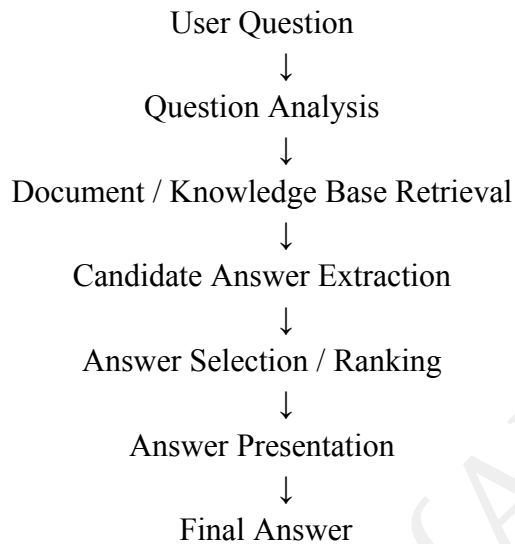
- The final answer is **presented to the user in natural language**.
- Some systems also provide a **context snippet** from the source document for verification.

Example:

Output: “Google was founded on September 4, 1998.”



Simplified Workflow Diagram



Example of Complete QAS Flow

Query: “Who invented the telephone?”

Question Analysis → Keywords: [invented, telephone], Answer type: Person

Document Retrieval → Document: “Alexander Graham Bell invented the telephone in 1876.”

Candidate Extraction → “Alexander Graham Bell”

Answer Selection → Highest relevance score: “Alexander Graham Bell”

Answer Presentation → Output: “Alexander Graham Bell”